

LUIS MATIAS

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Skills/Tools SolidWorks, Excel, Ansys, 3D Printing, Abaqus, Matlab, Python, Orca Slicer
Languages Portuguese (Native), English (C1 – Advanced), Spanish (A1 – Elementary)
Interests Mechanical Design, Machine Design, Product Development, CNC Machines, 3D Printing Additive Manufacturing, CAD/CAM, Manufacturing Engineer, Automation

WORK

Forvia ○ **VIE Development Engineer/Manufacturing Engineer** Trnava, Slovakia **May 2023 – Dec 2024**

- Working in Instrument Panel programs: Citroën C3 and Opel Frontera.
- Define and articulate the Engineering Levels required for production, serving as a key liaison between R&D and the plant team to ensure seamless collaboration.
- Validate mockups for both interim and permanent solutions, ensuring functionality and compliance with design standards.
- Prepare production documents including Visual BOM and 3D models in PLM systems.
- Provide support to customers during aspect and functional design evaluations.
- Assist manufacturing in defining the sequence of assembly operations during the initial stages of the conveyor production process, ensuring efficiency and accuracy.
- Independently develop 3D printing solutions for mockups, workstation and conveyor modifications, and production support tools, enhancing functionality and efficiency.
- Develop Excel VBA scripts to streamline stock tracking and ensure accurate inventory before SAP deployment.

Forvia ○ **VIE Program Engineer** Gorzów Wielkopolski, Poland **Aug 2022 – May 2023**

- Working in Door Panel programs: Jeep Avenger, Fiat 600 and Alfa Romeo Junior.
- Support quality department in capabilities studies.
- Support suppliers in Gage R&R studies to ensure RPS system definition.
- Support in the Meisterbock build.

INEGI ○ **Research Fellow** Porto, Portugal **Dec 2021 – July 2022**

Project: “Dealing with CT Cardiac Imaging Using Python: An Approach for Future Hemodynamic Simulations”

- Computer Vision - Using Python for 3D reconstruction of heart arteries. Region Growing method.
- User interface with PyQt.

Paper: “Dealing with CT Cardiac Imaging Using Python: An Approach for Future Hemodynamic Simulations”
Conference Paper. M2D2022 - 9th International Conference on Mechanics and Materials in Design (pp. 1109-1114)

STUDIES

University of Porto ○ **M.Sc. Mechanical Engineer** Porto, Portugal **July 2022**

Thesis: "Semi-Automatic Method of Stent Development in Coronary Arteries with Stenosis - Hemodynamic Simulations"

- Semi-automatic method to create arteries with stents using SolidWorks and Creo based on CT images.
- Hemodynamics simulations in ANSYS Fluent with stent applied in the arteries.

Paper: "Semi-Automatic Method of Stent Development for Hemodynamic Simulations in Patient Coronary Arteries with Disease"
Conference Paper. M2D2022 - 9th International Conference on Mechanics and Materials in Design (pp. 1109-1114)

MY PROJECTS

CNC Milling Machine **March 2025 – Now**

- Under development in design phase.
- Designing in SolidWorks.
- Ball Screw & Linear Guide driven system. Nema 23 stepper motors.
- T-Slot 40 Profile frame.

CNC Diode Laser/Engraver **Jan 2025 – March 2025**

- Finished project and working machine.
- Designed in SolidWorks.
- 3 Axis with autofocus. Air Assist.
- Pinion and Belt driven system with NEMA 17 stepper motors.
- V-Slot 2020 profile for frame and wheel rails. 3D Printing parts for feet and gate plates.

EXPERIENCES

Elephant Nature Park ○ **Volunteering | Elephant Caretaker** Chiang Mai, Thailand **2017 (1 Week)**

- Taking care of abandoned and abused elephants and dogs.

RMUTR ○ **Internship | Professor Assistant** Phutthamonthon, Thailand **2017 (3 Months)**

- Support students in their school projects.
- Teach Mathematics and Solid Mechanics to the students.

CERTIFICATIONS

[CSWP] SolidWorks CAD Design Professional **May 2025**
Dassault Systèmes

[CSWP-SU] SolidWorks Surfacing Professional **May 2025**
Dassault Systèmes